

23W-MATH-115A-LEC-3 Final!

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TOTAL POINTS

60 / 60

QUESTION 1

1 1a-e 10 / 10

- ✓ - 0 pts Correct
- 2 pts (a) wrong
- 2 pts (b) wrong
- 2 pts (c) wrong
- 2 pts (d) wrong
- 2 pts (e) wrong

QUESTION 2

2 1f-j 10 / 10

- ✓ + 2 pts Part (f) Correct
- ✓ + 2 pts Part (g) Correct
- ✓ + 2 pts Part (h) Correct
- ✓ + 2 pts Part (i) Correct
- ✓ + 2 pts Part (j) Correct
- + 0 pts None Correct

QUESTION 3

3 2 10 / 10

- ✓ - 0 pts Correct
- 3 pts major notational issues
- 4 pts failed to specify basis of eigenvectors
- 5 pts right idea, but major gaps in work
- 5 pts right ideas, but major error
- 10 pts completely wrong

QUESTION 4

4 3 10 / 10

- ✓ - 0 pts Correct
- 1 pts minor error
- 3 pts major issue with showing $S_1(\beta)$ is linearly independent
- 10 pts completely wrong
- 1 why is this so much darker than the rest??

QUESTION 5

5 4 10 / 10

- ✓ + 5 pts Forward Direction (Starting with $\ker(T)=\text{im}(T)$) Correct
- ✓ + 5 pts Reverse Direction (Starting with Even Dimension) Correct
- + 0 pts Incorrect

QUESTION 6

6 5 10 / 10

- ✓ + 5 pts Showed V is a sum of these subspaces.
- ✓ + 5 pts Showed trivial intersection.
- + 0 pts Incorrect

1. Please write "True" or "False" under each prompt. If we cannot distinguish what you wrote, we will treat it as incorrect. Each problem is worth 2 points, for a total of 20 points.

- (a) Let $T : V \rightarrow V$ be a linear operator on V , and let $W \subseteq V$ be T -invariant. W is necessarily T^5 -invariant as well.

TRUE

T

$w \in W, T(w) \in W, T(T(w)) \in W \dots$ ✓

- (b) A spanning set for a vector space V must be linearly dependent.

FALSE

F

basis

- (c) Let V be a vector space, and let $S \subseteq V$ be linearly independent. We can find a basis β for V such that $S \subseteq \beta$.

TRUE

T

- (d) If $T : V \rightarrow V$ where V is infinite dimensional, then we can find T that is onto but not one-to-one. TRUE

T

ex: deriv. map. $P(\mathbb{R}) \rightarrow P(\mathbb{R})$

$1 \rightarrow 0$
 $2 \rightarrow 0$

$ax^n + bx^{n-1} \mapsto nax^{n-1} + bx^{n-2}$ (n-1)

onto, not

$P(\mathbb{R})$ find:

use integral!

- (e) Let V be an inner product space, with inner product $\langle \cdot, \cdot \rangle$. We can complete orthonormal sets to orthonormal bases.

TRUE

T

- (f) Let $T : V \rightarrow V$ be a linear operator, and suppose $\dim(V) = n$. We can always find n distinct eigenvalues.

F

FALSE

$$\text{Id}_V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\mathbb{R}^2$

$$\dim(V) = 2$$

$$\lambda_1, \lambda_2 = 1$$

$\lambda_1 = \lambda_2$, ONE DISTINCT.

- (g) Let $T : V \rightarrow W$ be a linear map, and suppose $\dim(V) < \dim(W)$. We can find T that is one-to-one.

$$v \mapsto \dots \mapsto w$$

$v \nmid$ CAN BE SURJ, BUT inj.

TRUE

T

$$\begin{array}{ccc} V & \xrightarrow{\quad} & W \\ \downarrow & & \downarrow \\ 0 & \xrightarrow{\quad} & 0 \end{array}$$

$\dim(V) = 0 \quad \dim(W) = 1$

- (h) Let $T : V \rightarrow W$ be a linear map and suppose $\dim(V) < \dim(W)$. We can find T that is onto.

F

FALSE

- (i) Let $T : V \rightarrow V$ be a linear operator for a finite-dimensional space, and let λ be an eigenvalue. We must have $GM_T(\lambda) \leq AM_T(\lambda)$. That is, the geometric multiplicity is at most the algebraic multiplicity.

TRUE

T

$$1 \leq GM_T(\lambda) \leq AM_T(\lambda) \quad \checkmark$$

- (j) Let V be an inner product space, with inner product $\langle \cdot, \cdot \rangle$. An orthonormal set must be a basis.

F

FALSE

2. Let $T : V \rightarrow V$ be a linear operator on a finite-dimensional vector space. Suppose T has only one (distinct) eigenvalue. Suppose further that T is diagonalizable. Show that T is a scalar multiple of the identity. (Recall that the identity transformation is given by $Id_V : V \rightarrow V$, where $Id_V(v) = v$ for all $v \in V$.) [10 points]

Since T is diagonalizable, There exists a basis β_V for V that consists of eigenvectors associated with T 's eigenvalues. Let β_V be:

$$\beta_V = \{v_1, \dots, v_n\} \quad \text{and, for each } i \in \{1 \leq i \leq n\},$$

$$T(v_i) = \lambda_i v_i \quad \text{given associated eigenvalues } \lambda_1, \dots, \lambda_n. \quad \text{Since}$$

T only has one distinct eigenvalue, $\lambda_1 = \dots = \lambda_n$ and so

$$T(v_i) = \lambda_1 v_i. \quad \text{We can write that}$$

$$T(v_i) = 0v_1 + \dots + \lambda_1 v_i + \dots + 0v_n, \quad \text{implying}$$

$$[T(v_i)]_{\beta} = \begin{bmatrix} 0 \\ \vdots \\ \lambda_1 \\ \vdots \\ 0 \end{bmatrix} = e_i^n, \quad \text{where the } i^{\text{th}} \text{ row has entry } \lambda_1 \text{ and all other entries are zero.}$$

$$\text{Then, } [T]_{\beta} = \begin{bmatrix} [T(v_1)]_{\beta} & \dots & [T(v_n)]_{\beta} \end{bmatrix} = \lambda_1 \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 \end{bmatrix}$$

$$= \lambda_1 \cdot [Id_V]_{\beta} = \lambda_1 \cdot I_n, \quad \text{Identity matrix. } ([I_n]_{\beta} = I_n) \quad \text{for any basis } \beta$$

Hence, $T(v_i) = \lambda_1 (Id_V)(v_i)$ and T & $\lambda_1 (Id_V)$ agree on a basis

β_V , so $T = \lambda_1 (Id_V)$ and T is a scalar multiple of the identity transformation $Id_V : V \rightarrow V$ $\square$

Feel free to use this page for continuing problem 2

1e: assume all $c_i \in \mathbb{F}$, some field $\mathbb{F}$.

$$\dim(V) \leq \dim(W)$$

3. Let V, W, Z be finite-dimensional vector spaces. Suppose $S_1 : V \rightarrow W$ be a one-to-one linear map, and let $T : V \rightarrow Z$ be a linear map. Show that we can find $S_2 : W \rightarrow Z$ such that $S_2 \circ S_1 = T$. [10 points]

[Note: This was on your second midterm...]

Let $\beta_V = \{v_1, \dots, v_n\}$ be a basis for V . Consider the set $S_1(\beta_V) = \{S_1(v_1), \dots, S_1(v_n)\}$. WTS: $S_1(\beta_V)$ is linearly independent.

take some ^{arbitrary} linear combination of $S_1(\beta_V)$: $c_1 S_1(v_1) + \dots + c_n S_1(v_n) = 0$ (s.t. it equals zero). By linearity of S_1 , we know

$$c_1 S_1(v_1) + \dots + c_n S_1(v_n) = S_1(c_1 v_1 + \dots + c_n v_n) = 0.$$

This implies $c_1 v_1 + \dots + c_n v_n = 0$ because, by S_1 's injectivity,

$\ker(S_1) = \{0\}$. This is a lin. combination of vectors

of a basis for V , β_V , so $c_1 = \dots = c_n = 0$. Hence,

$S_1(\beta_V) = \{S_1(v_1), \dots, S_1(v_n)\}$ is linearly independent.

Therefore, we can extend $S_1(\beta_V)$ into a basis for W :

$$\begin{pmatrix} \dim V \leq \dim W \\ \Rightarrow \\ |\beta_V| \leq |\beta_W| \\ |\beta_V| = |\beta_V| \end{pmatrix}$$

Let $\beta_W = \{S_1(v_1), \dots, S_1(v_n), w_{n+1}, \dots, w_m\}$. We can construct

a linear map $S_2 : W \rightarrow Z$ such that $S_2(S_1(v_i)) = T(v_i)$ for

all $i \in \{1 \leq i \leq n\}$ and $S_2(w_j) = \vec{0}$ for all $j \in \{n < j \leq m\}$.

This maps all vectors of a basis of W , so S_2 is well-defined.

Since $S_2(S_1(v_i)) = [S_2 \circ S_1](v_i) = T(v_i)$ over all elements of the basis β_V , $S_2 \circ S_1$ and T agree on a basis!

HENCE: $[S_2 \circ S_1 = T]$ and we can find

such an S_2 .



Feel free to use this page for continuing problem 3

Note: assume $c_i, k_i \in F$ for some field F .

4. Let V be a finite-dimensional vector space. Show that $\dim(V)$ is even if and only if we can find $T: V \rightarrow V$ such that $\ker(T) = \text{Im}(T)$. [10 points]

[hint 1: one direction was from a past midterm...]

[hint 2: for the other direction, write down a basis. be smart about how you name/label your things.]

$\Leftarrow$ Suppose $\exists T: V \rightarrow V$ s.t. $\ker(T) = \text{Im}(T)$. That implies $\dim(\ker(T)) = \dim(\text{Im}(T)) = n$, for some $n \in \mathbb{Z}^{\geq 0}$, which holds since V is finite-dimensional. By Rank-nullity:

$$\dim(\ker(T)) + \dim(\text{Im}(T)) = \dim(V)$$

and so

$\dim(V) = n + n = 2n$, $n \in \mathbb{Z}^{\geq 0}$, in the form of an even number. Hence: $\dim(V)$ is even! $\square$

$\Rightarrow$ Suppose $\dim(V)$ is even. That means we can find an even-sized (has even cardinality) basis for V , β_V s.t.

$\beta_V = \{v_1, \dots, v_n, w_1, \dots, w_n\}$. Suppose we have a linear map

$T: V \rightarrow V$ s.t. $T(v_i) = 0 \quad \forall \{1 \leq i \leq n\}$ and

$T(w_i) = v_i \quad \forall \{1 \leq i \leq n\}$. The set $\{v_1, \dots, v_n\}$ is linearly independent. (since it's part of a basis).

Take arbitrary $v \in \text{span}\{v_1, \dots, v_n\}$. v can be written as $c_1 v_1 + \dots + c_n v_n$, and

$$T(v) = T(c_1 v_1 + \dots + c_n v_n) = c_1 T(v_1) + \dots + c_n T(v_n) = c_1(0) + \dots + c_n(0) = 0.$$

Hence, $v \in \ker(T)$. Since v was arbitrary, this shows that $\text{span}\{v_1, \dots, v_n\} \subseteq \ker T$.

To show $\ker(T) \subseteq \text{span}\{v_1, \dots, v_n\}$, take some $x \in V$ s.t.

$T(x) = 0$. we can write $x = c_1 v_1 + \dots + c_n v_n + k_1 w_1 + \dots + k_n w_n \Rightarrow$

$$T(x) = T(c_1 v_1 + \dots + c_n v_n) + T(k_1 w_1 + \dots + k_n w_n) = 0$$

we showed that $\forall v \in \text{span}\{v_1, \dots, v_n\}, T(v) = 0$. so,

$$T(x) = T(k_1 w_1 + \dots + k_n w_n) = 0 = k_1 T(w_1) + \dots + k_n T(w_n).$$

$T(x) = k_1 v_1 + \dots + k_n v_n = 0$. All $v_i \neq 0$ since they belong to a basis, so ...

Feel free to use this page for continuing problem 4

$T(x) = k_1 v_1 + \dots + k_n v_n = \emptyset$. This is a linear comb. of vectors in the basis $\beta_V = \{v_1, \dots, v_n, w_1, \dots, w_n\}$. So,

$k_1 = \dots = k_n = \emptyset$. $x \in V$ was arbitrary, so if $x \in \ker(T)$,

we represent $x = c_1 v_1 + \dots + c_n v_n + \emptyset + \emptyset + \dots + \emptyset = c_1 v_1 + \dots + c_n v_n$, so

$x \in \text{span}\{v_1, \dots, v_n\}$. $\ker(T) \subseteq \text{span}\{v_1, \dots, v_n\}$. So,

$\ker(T) = \text{span}\{v_1, \dots, v_n\}$!

WTS: $\text{Im}(T) = \text{span}\{v_1, \dots, v_n\}$. Take some $T(x) \in \text{Im}(T)$ for some $x \in V$ (by Image definition). We can write

$T(x) = T(c_1 v_1 + \dots + c_n v_n + k_1 w_1 + \dots + k_n w_n)$. As shown before,

$T(x) = \emptyset + k_1 v_1 + \dots + k_n v_n$, a lin. combination of $\{v_1, \dots, v_n\}$.

hence, $T(x) \in \text{span}\{v_1, \dots, v_n\} \Rightarrow \text{Im}(T) \subseteq \text{span}\{v_1, \dots, v_n\}$.

To show $\text{span}\{v_1, \dots, v_n\} \subseteq \text{Im}(T)$, take $x \in \text{span}\{v_1, \dots, v_n\}$.

$x = c_1 v_1 + \dots + c_n v_n$ WTS: $x \in \text{Im}(T)$.

$\text{Im}(T) = \{w \in T(v) \mid v \in V\}$. $v = c_1 v_1 + \dots + c_n v_n + k_1 w_1 + \dots + k_n w_n$ for some $c_i, k_i \in F$.

write $T(v) = T(c_1 v_1 + \dots + c_n v_n + k_1 w_1 + \dots + k_n w_n) = w$

$= k_1 v_1 + \dots + k_n v_n$. Any $w \in T(v)$ takes on this form,

and $\text{Im}(T)$ is a subspace, so any LIN. combination of $\{v_1, \dots, v_n\}$ must

belong to $\text{Im}(T)$. Hence, $x \in \text{Im}(T)$ and x was arbitrary $\in \text{span}\{v_1, \dots, v_n\}$.

So $\text{span}\{v_1, \dots, v_n\} \subseteq \text{Im}(T) \Rightarrow \text{Im}(T) = \text{span}\{v_1, \dots, v_n\}$.

Since $\ker(T) = \text{span}\{v_1, \dots, v_n\} = \text{Im}(T)$, we found T s.t.

$[\ker(T) = \text{Im}(T)]$ given $\dim(V)$ is even. □

BOTH $\Leftarrow$ & $\Rightarrow$ have been proven!
directions

5. Let $T : V \rightarrow V$ be a linear operator on a finite-dimensional vector space. Suppose further that $T^2 = T$. Show that $V = \ker(T) \oplus \text{Im}(T)$. [10 points]

[hint 1: you may use the following dimension theorem without proof: Suppose $W_1, W_2 \subseteq V$ are subspaces of a finite-dimensional vector space. Then we have $\dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2) = \dim(W_1 + W_2)$.]

[hint 2: there are multiple ways to do this.]

Take an arbitrary element $x \in \ker(T) \cap \text{Im}(T)$. WTS: $x = 0$.

Since $x \in \ker(T)$, $T(x) = 0$. Since $T^2 = T$, $T(x) = T(T(x)) = 0$.
that implies $T(x) \in \ker(T)$ as well.

Since $x \in \text{Im}(T)$, we can find $v \in V$ such that $T(v) = x$.

By $T^2 = T$, that implies $T(v) = T(T(v)) = x$.

similarly, we can transform both sides of $T(v) = x \Rightarrow T(T(v)) = T(x)$.

therefore, $T(T(v)) = T(x) = x$. But since $x \in \ker(T)$,

$T(x) = 0$, so $T(T(v)) = 0 = x$. x MUST be zero! x was arbitrarily chosen from $\ker(T) \cap \text{Im}(T)$.

Hence, any element of $\ker(T) \cap \text{Im}(T)$ is zero.

$$\ker(T) \cap \text{Im}(T) = \{0\}. \quad \underline{\dim(\ker T \cap \text{Im} T) = 0.}$$

this is zero.

(hint 1)

$$\dim(\ker(T)) + \dim(\text{Im}(T)) - \overbrace{\dim(\ker T \cap \text{Im} T)}^{\text{this is zero}} = \dim(\ker T + \text{Im} T)$$

$$\Rightarrow \text{null}(T) + \text{rank}(T) - 0 = \dim(\ker T + \text{Im} T).$$

~~Since~~ By Rank-Nullity, $\text{null}(T) + \text{rank}(T) = \dim(V)$, so

$$\dim V = \dim(\ker T + \text{Im} T).$$

Because $\ker(T) \subseteq V$ and $\text{Im}(T) \subseteq V$ are SUBSPACES of V , we can take some $v_1 \in \ker(T)$ and $v_2 \in \text{Im}(T)$. $v_1, v_2 \in V$ by subspace property, and by ~~the~~ property of vector spaces,

$v_1 + v_2 \in V$. This defines the sum of $\ker(T)$ & $\text{Im}(T)$:

$$\ker T + \text{Im} T = \{v_1 + v_2 \mid v_1 \in \ker T, v_2 \in \text{Im} T\}, \text{ so}$$

$\ker(T) + \text{Im}(T) \subseteq V$. $\ker T + \text{Im} T$ is a SUBSPACE of V and

$\dim[\ker T + \text{Im} T] = \dim V$, so it MUST hold that $\ker T + \text{Im} T = V$.

Feel free to use this page for continuing problem 5

We have established that

$$\textcircled{1} \quad \ker(T) + \operatorname{Im}(T) = V \quad \text{and}$$

$$\textcircled{2} \quad \ker(T) \cap \operatorname{Im}(T) = \{\emptyset\}, \quad \text{s.o.}$$

$$\ker(T) \oplus \operatorname{Im}(T) = V.$$



~~(*) aside: Proof for if $W \subseteq V$, a subspace W of V , and $\dim(W) = \dim(V)$,
then $W = V$.~~